Kanan Aliyev

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EDUCATION

Northeastern University *Expected Dec 2026*
Master of Science in Computer Science Seattle, WA

Lawrence University *Sep 2020 – Jun 2024*
Bachelor of Arts in Computer Science Appleton, WI

PUBLICATIONS & MANUSCRIPTS

K. Aliyev and N. Kalavakonda, "Persistent Scene-Graph Memory on a Legged Robot: Separating Memory Gains from Execution Failure," Conference on Robot Learning (CoRL) 2026, Late Breaking Results. Also accepted at the CoRL 2026 Workshop on Memory for Robot Foundation Models (RoboMem).

RESEARCH

Persistent Scene-Graph Memory for Object-Goal Navigation on a Legged Robot *Jun 2026 – Present*

- Built a Go2 navigation system in **Isaac Sim** that maps rooms, places and objects into a live 3D scene graph (**Hydra**) and takes open-vocabulary requests
- Designed a persistent object memory: when asked again for an object it has already seen, the robot goes straight to the remembered location instead of searching again, even after the map reorganizes itself
- Ran the evaluation with a failure analysis that separates failures in perception and planning from failures in walking (falls and stalls), so gains from memory are not confused with locomotion errors
- Trained a custom stair-climbing policy, tested in simulation only; real stair runs use Unitree's built-in controller

EXPERIENCE

YSAP Inc., *AI Engineering Intern* *Sep 2025 – Jan 2026*

- Built and deployed a production retrieval-augmented (RAG) search over **40,000+** sounds, presets and devices, answering queries in under **2** seconds
- Built a UI automation tool that maps an application's full interface in under **10** seconds, so software can find, highlight and operate any on-screen element
- Built the audio-similarity search and cataloging pipelines behind a lesson-building tool for tutors, combining vector embeddings with a relational database

Osgil Defense, *Graduate Researcher* *May 2024 – Aug 2024*

- Built and deployed a system that rates how trustworthy an LLM's answers are: it searches the web in real time to fact-check claims and verify the cited sources
- Built the source-tracking pipeline behind it: parallel web scraping, a unique ID per source, and tagged citations carried through to the trust score
- Streamed LLM responses over **WebSockets**, summarized long sources to cut cost, and deployed it on GPUs

PROJECTS

Programmable-Lidar Navigation Testbed (Industry Collaboration) *Sep 2026 – Present*

- Leading a team of **6** comparing autonomous navigation in office and warehouse-style spaces between two **TurtleBot 4** robots, one with the stock sensors and one adding an industry partner's programmable solid-state lidar; main technical contact with the partner's engineers

Go1 Voice Navigation (MS Project) *Jan 2026 – Apr 2026*

- Built voice-controlled navigation for a Unitree Go1: **Whisper** transcribes the command, an LLM turns it into navigation calls, and **Nav2** drives the robot on a lidar SLAM map; a web dashboard shows live camera and lidar feeds and takes click-to-navigate goals
- Split the system between an onboard computer and a remote GPU server for speech recognition, and bridged the lidar's **ROS 1** driver into **ROS 2**

TARS: LLM Agents for Penetration Testing *Apr 2024 – May 2024*

- Built a multi-agent LLM system that automates penetration testing: agents drive standard security tools (**Nmap, Metasploit, Burp Suite, SQLMap**) to scan for vulnerabilities and analyze threats; about **350** GitHub stars
- Gave the agents a sandboxed shell tool to run the security tools inside isolated **Docker** containers

Custom JetRacer (Senior Capstone) *Sep 2023 – Dec 2023*

- Built an autonomous racing car on an **Nvidia Jetson** that follows the road from camera images, running the vision model at **65** FPS
- Built the training loop around it: camera capture, click-based labeling, and live evaluation of each new model

SKILLS

Robotics & Perception: ROS 2, Nav2, MoveIt 2, Isaac Sim / Isaac Lab, 3D Scene Graphs (Hydra), Lidar SLAM (2D and 3D), Occupancy Mapping, RGB-D & Point-Cloud Data, Open-Vocabulary Grounding, Crowd Simulation, TurtleBot 4, Programmable Solid-State Lidar, Reinforcement Learning for Locomotion

ML & LLM Systems: PyTorch, ONNX Runtime, TensorRT, OpenCV, RAG & Vector Search (FAISS), LLM Agents & Tool Use (LangChain, CrewAI)

Languages & Tools: Python, C++, Java, C# / .NET, React, SQLite, Linux, Docker, Git, GCP (Vertex AI), AWS (Lambda, S3, EC2), Nvidia Jetson, CAD, 3D Printing